FILE PERMISSIONS IN LINUX

Tools of the Trade: Linux and SQL – Manage Authorization

Project description:

My organization's research team required updates to the file permissions for specific files and directories located in the projects directory. The current permissions did not align with the necessary authorization levels. By reviewing and modifying these permissions, I aimed to maintain system security. To accomplish this objective, I carried out the following steps:

Check file and directory details

The following code illustrates how I used Linux commands to examine the existing permissions for a particular directory within the file system.

The initial line of the screenshot shows the command I input, while the subsequent lines display the resulting output. The code lists all contents within the projects directory. I applied the ls command with the -la option to generate a detailed listing of file contents that also includes hidden files. The command output reveals one directory named drafts, one hidden file called .project_x.txt, and five additional project files. The 10-character string appearing in the first column represents the permission settings assigned to each file or directory.

```
researcher2@c1f9ca8ac39f:~$ pwd
/home/researcher2
researcher2@c1f9ca8ac39f:~$ ls
projects
researcher2@c1f9ca8ac39f:~$ cd /home/researcher2/projects
researcher2@c1f9ca8ac39f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 13 02:44 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 13 04:58 ..
-rw--w---- 1 researcher2 research_team   46 Mar 13 02:44 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 13 02:44 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Mar 13 02:44 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar 13 02:44 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_t.txt
researcher2@c1f9ca8ac39f:~/projects$
```

Describe the permissions string

The 10-character string can be broken down to identify who has access to each file and what specific permissions they hold. The meaning of each character position is as follows:

- **1st character:** This character is either a d or a hyphen (-) and indicates the file type. A d signifies a directory, while a hyphen (-) denotes a regular file.
- **2nd-4th characters:** These characters show the read (r), write (w), and execute (x) permissions for the user. If a hyphen (-) appears in place of any of these letters, it means that particular permission is not granted to the user.
- **5th-7th characters:** These characters display the read (r), write (w), and execute (x) permissions for the group. A hyphen (-) in any position indicates that permission is not given to the group.
- **8th-10th characters:** These characters reveal the read (r), write (w), and execute (x) permissions for other users. This category includes all users on the system except the user and the group. A hyphen (-) in any spot means that permission is not granted to others.

For instance, the file permissions for project_t.txt are -rw-rw-r--. Because the first character is a hyphen (-), this confirms that project_t.txt is a file rather than a directory. The second, fifth, and eighth characters are all r, indicating that user, group, and others all possess read permissions. The third and sixth characters are w, showing that only the user and group have write permissions. No execute permissions are assigned to anyone for project_t.txt.

Change file permissions

The organization determined that other users should not have write access to any files. To follow this requirement, I reviewed the file permissions I had previously obtained. I identified that project_k.txt needed write access removed for others.

The following code demonstrates how I used Linux commands to implement this change:

The first two lines of the screenshot display the commands I entered, and the remaining lines show the output from the second command. The chmod command modifies permissions on files and directories. The first argument specifies which permissions to change, while the second argument identifies the target file or directory. In this case, I removed write permissions for others from the project_k.txt file. Following this change, I used ls -la to verify the updates I had made.

```
researcher2@c1f9ca8ac39f:~/projects$ chmod o-w project_k.txt
researcher2@c1f9ca8ac39f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 13 02:44 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 13 04:58 ..
-rw--w---- 1 researcher2 research_team   46 Mar 13 02:44 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 13 02:44 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar 13 02:44 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_t.txt
researcher2@c1f9ca8ac39f:~/projects$
```

Change file permissions on a hidden file

The research team recently archived project_x.txt. They decided that no one should have write access to this project, although the user and group should retain read access. The following code illustrates how I used Linux commands to adjust these permissions:

The first two lines of the screenshot show the commands I entered, and the other lines display the output from the second command. I recognized .project_x.txt as a hidden file because it begins with a period (.). In this example, I removed write permissions from both the user and group, while also adding read permissions to the group. Specifically, I used u-w to take away write access from the user, g-w to remove write access from the group, and g+r to grant read access to the group.

```
researcher2@c1f9ca8ac39f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 13 02:44 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 13 04:58 ..
-rw--w---- 1 researcher2 research_team   46 Mar 13 02:44 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 13 02:44 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_k.txt
-rw------- 1 researcher2 research_team   46 Mar 13 02:44 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_t.txt
researcher2@c1f9ca8ac39f:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@c1f9ca8ac39f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 13 02:44 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 13 04:58 ..
-r--r----- 1 researcher2 research_team   46 Mar 13 02:44 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 13 02:44 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_k.txt
-rw------- 1 researcher2 research_team   46 Mar 13 02:44 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_t.txt
researcher2@c1f9ca8ac39f:~/projects$
```

Change directory permissions

My organization determined that only the user researcher2 should have access to the drafts directory and its contents. This means that no one besides researcher2 should possess execute permissions.

The following code demonstrates how I used Linux commands to apply this change:

The output displays the permission listings for multiple files and directories. Line 1 shows the current directory (projects), and line 2 indicates the parent directory (home). Line 3 represents a regular file named .project_x.txt. Line 4 is the drafts directory with modified permissions. Here, only researcher2 has execute permissions. After initially finding that the group had execute permissions, I used the chmod command to revoke them. Since researcher2 already had execute permissions, no additional changes were necessary for that user.

```
researcher2@c1f9ca8ac39f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 13 02:44 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 13 04:58 ..
-r--r----- 1 researcher2 research_team   46 Mar 13 02:44 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 13 02:44 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_k.txt
-rw------- 1 researcher2 research_team   46 Mar 13 02:44 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_t.txt
researcher2@c1f9ca8ac39f:~/projects$ chmod g-x drafts
researcher2@c1f9ca8ac39f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 13 02:44 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 13 04:58 ..
-r--r----- 1 researcher2 research_team   46 Mar 13 02:44 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Mar 13 02:44 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_k.txt
-rw------- 1 researcher2 research_team   46 Mar 13 02:44 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 13 02:44 project_t.txt
researcher2@c1f9ca8ac39f:~/projects$
```

Summary

I adjusted several permissions to meet my organization's authorization requirements for files and directories within the projects directory. I began by running ls -la to examine the current permission settings, which guided my subsequent decisions. Then, I applied the chmod command multiple times to modify permissions on the appropriate files and directories.